



Using Numeric Data Types

Practical Skills Module 2: Topic 2

[Visit our website](#)

Practical Skills Module 2: Topic 2

Topic Code	PM-02-PS02:
Topic	Use numbers in Python

Scope of Practical Skill

Given an applicable instruction and access to a learning platform, the learner must be able to:

- PA0201 Create an integer, through assigning the integer value in a variable
- PA0202 Create a number with the float data type, through assigning the value into a variable
- PA0203 Check the data type of a variable or data using python's built-in `type()` function
- PA0204 Create large numbers in python using the underscore (`_`)
- PA0205 Add, subtract, multiply, divide

Applied Knowledge

- AK0201 Concept, definition and functions of Python programming functionalities

Internal Assessment Criteria

- IAC0201 Expected results with numbers in Python are achieved

Resources:

Knowledge	Information from Python Programmer Curriculum
National Curriculum Framework	900221-000-00-PM-02



Practical task

Follow the facilitator's instructions to complete the following activities:

Scenario: Building a Simple "Shopping Cart Calculator"

In this scenario, you are building a program to manage and calculate prices in a shopping cart. The steps involve working with integers, floats, and arithmetic operations to compute the total price, tax, and discounts. This program will simulate a basic price calculation system used in online stores.

Create Python file called **online_store.py** and complete the following:

Step 1: Assign an integer value to represent quantity.

- Assign an integer value to a variable that represents the **quantity of items** in the shopping cart.
- Use a clear and descriptive variable name (e.g., `item_quantity`).

Step 2: Assign a float value to represent price.

- Assign a float value to a variable that represents the **price of a single item**.
- Ensure the value includes decimals to represent cents.

Step 3: Use the `type()` function to check data types.

- Use the `type()` function to check the data type of both variables created in Step 1 and Step 2.
- Print the result of each check.
 - *Expected output:* One result should indicate `<class 'int'>`, the other `<class 'float'>`.

Step 4: Assign a large number using underscores.

- Assign a large number (e.g., total shopping budget) to a variable using underscores for readability.
- Example: `1_000_000` for one million.
- Print the value to confirm it behaves like a regular number.
 - *Expected output:* The printed value should display without underscores (e.g., `1000000`).

Step 5: Perform arithmetic operations.

Use the variables created so far to complete the following:

- **Calculate the total cost** by multiplying the item quantity by the item price.
- **Apply a discount** by subtracting a fixed amount (you decide the value) from the total cost.
- **Calculate the tax amount** by applying a tax percentage (e.g., 15%) to the discounted total.
- **Compute the final amount payable** by adding the tax to the discounted total.
- Print the result of each calculation step with a label so it's easy to understand.

Expected output: Using the following example inputs:

- Item quantity: 3
- Item price: 249.99
- Discount: 50
- Tax rate: 0.15 (i.e., 15%)

Your output should look similar to the example below:

```
Total cost before discount: 749.97
Total after discount: 699.97
Tax amount: 104.9955
Final amount payable: 804.9655
```



Take note

Important: Be sure to upload all files required for the task submission inside your task folder and then click "Request review" on your dashboard.